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Conference Paper · December 2013

DOI: 10.1109/ICECS.2013.6815464

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Simulated Analysis and Hardware Implementation of Voiceband Circular Microphone Array

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***Abstract*—Acoustic microphone array beamforming represents an adequate technology for remote acoustic surveillance and multi conference recording, as the system has no mechanical parts and it has moderate size. This paper contains both the simulated analysis and hardware implementation of an 8-channel uniform circular array operating with fixed filter-and-sum (FaS) beamforming. The hardware implementation comprises essentially the analog and the digital parts. In the analog part, the array frame with high-sensitivity small electret microphones and low-noise JFET-based microphone preamplifiers were built. In the digital part, a multi-channel audio DSP board was used to sample the preamplifiers output and implement the real-time FaS digital processing. The real performance analysis of the prototype in medium-sized meeting rooms was successfully carried out.**

I. Introduction

Acoustic array beamforming [1]–[3] refers to the ability of a set of microphones at capturing a desired sound arriving from a certain spatial direction while rejecting undesired sounds coming from other directions. As the microphone array does not have mechanical movable parts, it can be used unobtrusively for acoustic surveillance or multiconferencing [4]. A microphone array is composed of several microphones aligned in a specific pattern or geometry to operate as a single device. Due to its compactness, the circular geometry for a moderate number of microphones is suitable in sound beamforming.

Microphone arrays appeared nearly three decades ago, but they still are a field of active research [5]–[10]. The simplest approach for combining the microphone signals is with the so-called delay-and-sum (DaS) beamforming. In wideband signals like speech (and audio in general), the performance delivered by DaS may not be sufficient. Nevertheless, the use of specific digital filters applied on every channel can substantially boost the global signal to noise/interference ratio (SNR). Interestingly, this filter-and-sum (FaS) beamforming can be seen as strategically placing a larger number of "virtual" microphones, hence improve the spatial discrimination.

The aim of this paper is to explore and understand the sound discrimination abilities of a uniform circular microphone array operating in the voice spectral band. With such an aim, the simulated performance analysis of DaS and FaS fixed beamforming algorithms, detailed in Sec. II, and the complete hardware implementation, in Sec. III, have been carried out; the conclusions and further research work close the paper.

This work has been partially supported by UAE ICT Fund. This work has been awarded the first prize in UAE 2013 IEEE Student Day.

II. Circular Microphone Array

Fig. 1 illustrates schematically the delay pattern in a microphone array with uniform circular geometry and direction of arrival (DoA) of the sound source equal to θ. In far-field conditions (distance to the sound source much larger than array dimensions) the signal at the ith microphone results in

$$x_i(t) = s(t - \delta_i) \tag{1}$$

where t is continuous time, and δ_i represents the physical time delay at the ith microphone with respect to the centre of the array. It is simple to deduce that for such a geometric arrangement, such a time delay follows the rule

$$\delta_i = (r/c)\cos\big(\theta - i2\pi/N\big) \tag{2}$$

where r is the radius of the circular geometry, c is the speed of sound in air, N is the number of microphones, and the microphone index runs as $i = 1, \cdots, N$. It is important to relate the radius r of the array with the distance between adjacent microphones d, a relation that follows

$$d = 2r\sin(\pi/N). \tag{3}$$

The output of the microphones (1) are then converted to discrete time n at a sampling frequency f_s, resulting in $x_i[n]$.

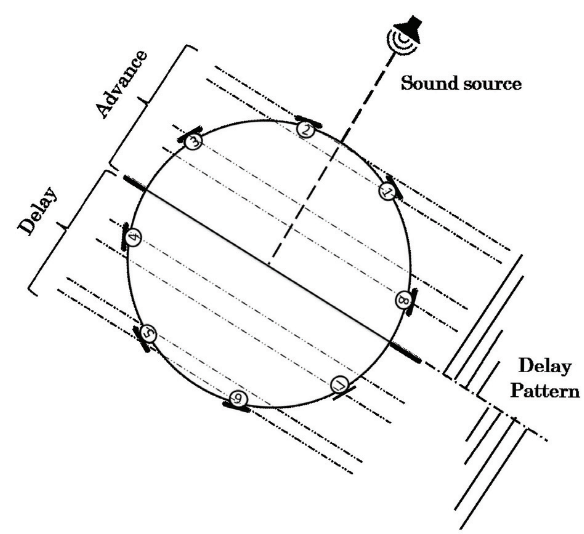

Fig. 1. Time of arrival on a 8-microphone uniform circular array.

A. Delay-and-Sum Beamforming

In the basic delay-and-sum (DaS) algorithm, the digital signal at the ith channel is simply (anti) delayed[1]

$$y_i[n] = x_i[n + \Delta_i] \tag{4}$$

where Δ_i corresponds to the digital delay. As the array is supposed to be "listening" to the steering angle ϕ, hence

$$\Delta_i = f_s(r/c)\cos\big(\phi - i2\pi/N\big). \tag{5}$$

Note that Δ_i is meant to be an integer value, which corresponds to a delay in discrete samples. The beamformer output is built with the (average) sum of all channels as follows

$$y[n] = \frac{1}{N}\sum_{i=1}^{N} y_i[n] = \frac{1}{N}\sum_{i=1}^{N} s[n - f_s\delta_i + \Delta_i] \tag{6}$$

where $s[n]$ is the discrete-time equivalent signal of the sound source. If the actual direction of the incoming sound θ happens to be close to the steering angle ϕ, the physical and digital delays compensate ($\Delta_i \simeq f_s\delta_i$), and the beamformer output results in the sound source signal undistorted. In general, the averaging among the signals results in the frequency-selective attenuation of the sound source, as the translation of the beamformer output (6) to the frequency domain reveals

$$Y(e^{j\omega}) = D(\theta,\omega)\, S(e^{j\omega}) \tag{7}$$

where ω denotes frequency $-\pi < \omega \leq \pi$. Here $Y(e^{j\omega})$ and $S(e^{j\omega})$ correspond to the Fourier transform of $y[n]$ and $s[n]$ respectively, and $D(\theta,\omega)$ is the transfer function of the DaS beamformer with steering angle ϕ

$$D(\theta,\omega) = \frac{1}{N}\sum_{i=1}^{N} \exp\big(-j\omega(f_s\delta_i - \Delta_i)\big). \tag{8}$$

B. Filter-and-Sum Beamforming

In the filter-and-sum (FaS) beamforming, each microphone signal is input to a finite-length impulse response (FIR) filter

$$y_i[n] = \sum_{k=-K}^{K} a_{i,k}\, x_i[n-k] \tag{9}$$

where K defines the time span of the FIR filter, and $\boldsymbol{a}_i = [a_{i,-K} \ \cdots a_{i,0} \ \cdots \ a_{i,K}]$ are the coefficients of the ith FIR filter. The total beamformer output is the sum of all channels

$$y[n] = \sum_{i=1}^{N} y_i[n]. \tag{10}$$

This digital signal processing (10) is illustrated in Fig. 2.

Similarly to (7), here

$$Y(e^{j\omega}) = F(\theta,\omega)\, S(e^{j\omega}) \tag{11}$$

where $F(\theta,\omega)$ is the transfer function of the FaS beamformer

$$F(\theta,\omega) = \sum_{i=1}^{N}\sum_{k=-K}^{K} a_{i,k} e^{-j\omega k} e^{-j\omega f_s \delta_i}. \tag{12}$$

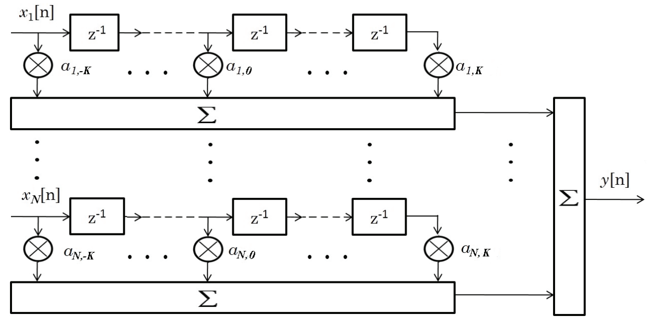

Fig. 2. Signal processing operation of filter-and-sum beamforming.

In order to prevent spectral distortion, $F(\theta,\omega)$ must depend only on the DoA θ, that is, to be equal to a desired beampattern $B(\theta)$ that is not a function of frequency ω. Therefore, we aim here to make minimum the mean square error (MSE)

$$\min_{\boldsymbol{a}} \iint \big|B(\theta-\phi) - F(\theta,\omega)\big|^2 d\omega\, d\theta \tag{13}$$

where $\boldsymbol{a} = \big[\boldsymbol{a}_1 \cdots \boldsymbol{a}_N\big]$ is the vector with all N FIR filter coefficients. Note that the steering angle ϕ simply appears in the form of a shift of the desired beampattern. The desired beampattern must obviously fulfil $B(0) \geqslant B(\theta)$ and $B(\pi) \approx 0$; this shape will determine the aspect of the beam and its width: a Gaussian shape has proven experimentally to yield a good compromise between beam width and attenuation.

We can rewrite the transfer function (12) as

$$F(\theta,\omega) = \sum_{i=1}^{N} \boldsymbol{a}_i\, \boldsymbol{z}_i^H(\theta,\omega) = \boldsymbol{a}\, \boldsymbol{z}^H(\theta,\omega) \tag{14}$$

where ^{H} denotes transpose and complex conjugate, and

$$\boldsymbol{z}_i(\theta,\omega) = \big[e^{-j\omega K} \ \cdots \ 1 \ \cdots \ e^{j\omega K}\big]\, e^{-j\omega f_s \delta_i} \tag{15a}$$
$$\boldsymbol{z}(\theta,\omega) = \big[\boldsymbol{z}_1(\theta,\omega) \ \cdots \ \boldsymbol{z}_N(\theta,\omega)\big]. \tag{15b}$$

Since the transfer function (14) corresponds to a linear combination of the FIR coefficients, the optimization problem (13) is easy to solve. The integral can be evaluated numerically

$$\min_{\boldsymbol{a}} \sum_{\ell=1}^{L}\sum_{m=1}^{M} \big|B(\theta_\ell - \phi) - \boldsymbol{a}\, \boldsymbol{z}^H(\theta_\ell,\omega_m)\big|^2 \tag{16}$$

where θ_ℓ and ω_m corresponds to a $L \times M$ grid of the DoA and frequency respectively. This allows us to define the desired beampattern vector $\boldsymbol{d}$ and the input at frequency bin m as

$$\boldsymbol{b}_\phi = \big[B(\theta_1 - \phi) \cdots B(\theta_L - \phi)\big] \tag{17}$$
$$\boldsymbol{Z}_m = \big[\boldsymbol{z}^H(\theta_1,\omega_m) \cdots \boldsymbol{z}^H(\theta_L,\omega_m)\big]\,. \tag{18}$$

The optimal FIR coefficients are finally obtained by a matrix inversion as follows

$$\boldsymbol{a} = \left(\sum_{m=1}^{M} \boldsymbol{Z}_m \boldsymbol{Z}_m^H\right)^{-1} \left(\boldsymbol{b}_\phi \sum_{m=1}^{M} \boldsymbol{Z}_m^H\right). \tag{19}$$

[1]A (pseudo) antidelay is practically implemented by buffering the discrete-time signal, thus having access to "future" samples.

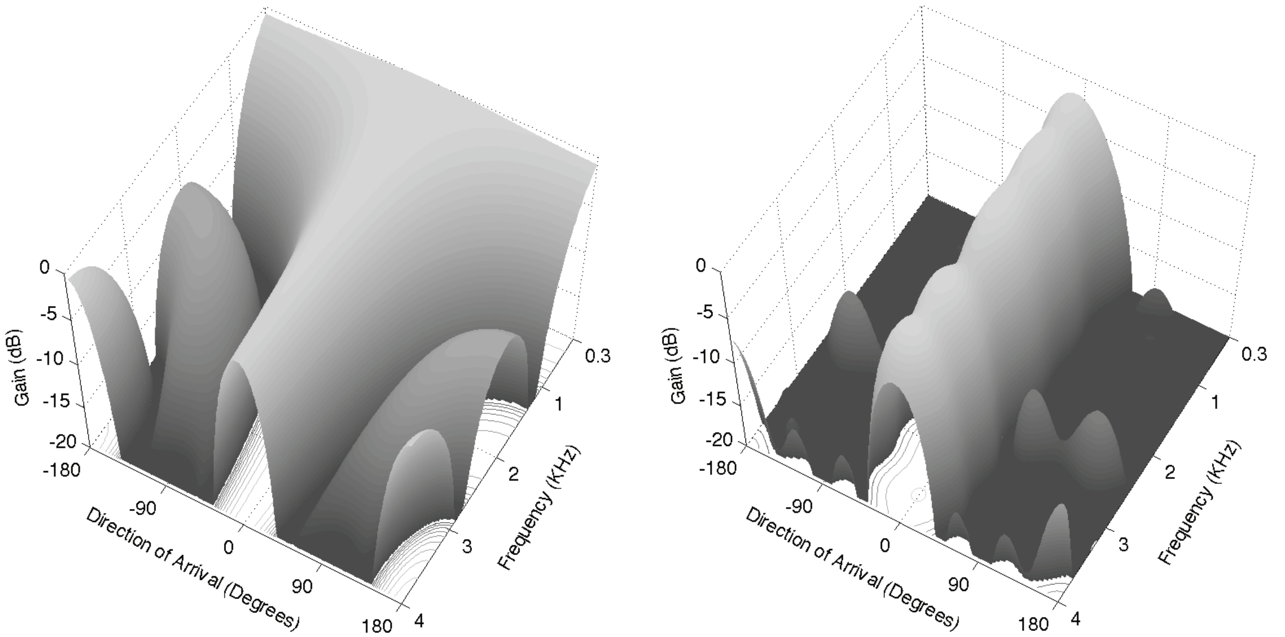

Fig. 3. Beam pattern of an 8-microphone circular array for: delay-and-sum (DaS) beamforming (left) and filter-and-sum (FaS) beamforming (right).

C. Simulated Comparative Analysis

The FaS beamforming possesses more degrees of freedom than the simple DaS processing. In fact this one can be seen as a particular case of the former, namely, an all-zero FIR except $a_{i,k} = 1/N$ for $k = f_s \delta_i$. In order to establish an objective comparison, we evaluated the beampattern $D(\theta, \omega)$ and $F(\theta, \omega)$ on the same scenario. The distance between microphones was set to $d = 4\,\text{cm}$; this value corresponds to half a wavelength of a $4\,\text{kHz}$ sound travelling on air; thus we expect to have an spatial-aliasing-free beampattern within the spectral band of human speech intelligibility (up to $4\,\text{kHz}$). For reasons detailed in the next section, the number of microphones was set to $N = 8$. The FIR time span in the FaS beamforming was $K = 19$ (a value selected experimentally).

Fig. 3 illustrates the beampattern delivered by DaS and FaS on the same array structure. The DaS processing yields a beampattern with high-level granting lobes and a non-uniform main beam, hence uncapable to deliver adequate sound spatial discrimination. On the contrary, the flexibility that $(2K + 1)$ FIR coefficients per channel deliver, makes the FaS algorithm adequate for sound spatial discrimination: the resulting beampattern is nearly frequency-independent, and very few spurious low-gain granting lobes exhibit. The polar plot of the beampattern at certain frequencies, shown in Fig. 4, illustrates further the benefits of FaS: a narrow beam that is nearly frequency independent.

In turn, the computational burden in the DaS is negligible, while the FaS requires $(2K + 1)N$ multiply-and-accumulate operations per sample according to a canonic FIR filter operation. In the scenario of interest, $K = 19$, $N = 8$, and $f_s = 16000$, the computational load becomes nearly $2.4\,\text{MOp/s}$, which can be accomplished with today's digital signal processors (DSP).

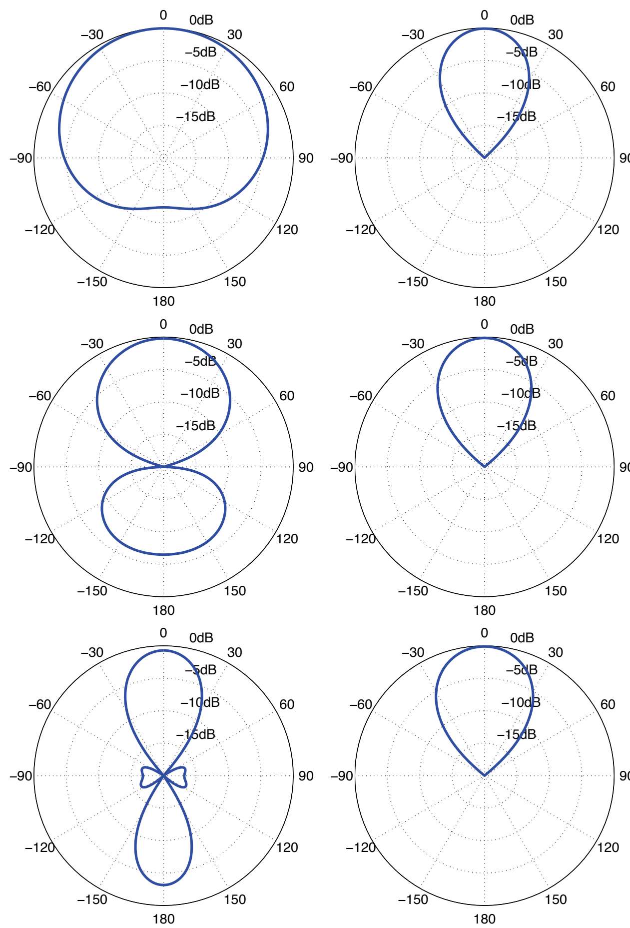

Fig. 4. Beam pattern polar plot for DaS (left) and FaS (right) for the following frequencies: $1\,\text{kHz}$ (top), $2\,\text{kHz}$ (middle), and $3.5\,\text{kHz}$ (bottom).

III. Hardware Implementation

Bearing the previous simulated analysis in mind, we built a hardware prototype of a uniform circular array, shown in Fig. 5. The system is composed of four main parts: the frame with 8 microphones, the microphone preamplifiers, the analog-digital converters (ADC), and the DSP computing board.

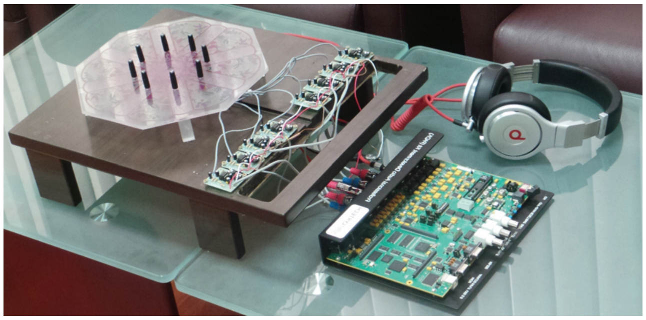

Fig. 5. 8-microphone uniform circular array prototype ($d = 4\,\mathrm{cm}$).

The microphones correspond to the model EKULIT EMY-4530M, a high-sensitivity ($-65\,\mathrm{dB}$) $0.15\,\mathrm{mA}$ omnidirectional electret microphone of flat frequency response within 50-15000 Hz. The microphone preamplifiers, shown in Fig. 6, are based on Texas Instrument LF351 model, a wideband audio low-noise JFET input operational amplifier; the resistor R was set accordingly to obtain a three-order-magnitude amplification gain of the microphone output. Finally, Lyrtech's professional audio development kit (PADK) provided the ADCs and DSP required to digitally process the output signals from the preamplifiers. The PADK consists of eight 24-bit AD (as well as DA) converters operating at a maximum sampling frequency of $192\,\mathrm{kHz}$, and the TMS320C6727 DSP of peak computing power equal to 2.4 GFlops. While the PADK performance exceeds by far the system requirements, it sets the maximum number of microphones in the array to eight.

As the main intended usage of the system is as directional microphone for conference and meetings held in a room or hall, we tested the system under those real conditions: the system was placed on a table with five different participants sitting around it. The steering angle was programmed to be set manually from the PADK switches and buttons. In listening tests it became clear that the person who was placed at the steering angle was recorded clearly and without noticeable distortion. On the other hand, the recording of other participants resulted in a blurred unintelligible conversation. However, those interfering sounds were not attenuated as predicted in our analysis concluded in Fig. 3. This mismatch can be explained from the two-dimensional (2D) model used in the analysis versus the 3D acoustic characteristics of the real scenario. That is, the sound reflections taking place in the room arrive to the array from elevation angles different than zero (the horizontal plane where the microphones are lying). We believe that this problem can be successfully addressed with a software-only solution, without any hardware modification. This research avenue is currently under investigation.

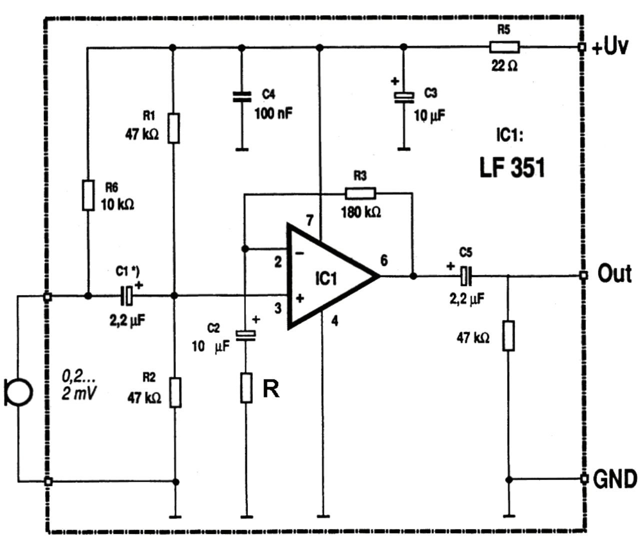

Fig. 6. Low-noise wideband audio microphone preamplifier ($U_V = 11\,\mathrm{V}$).

IV. Conclusions

In this work we have validated theoretically and practically the sound discrimination capabilities of a 8-channel uniform circular microphone array. For such an objective, the simulated analysis of the filter-and-sum (FaS) fixed beamforming along with the actual hardware implementation of the device were carried out. The validation of the prototype in real acoustic scenarios show manifestly the good performance of the system. However, the discrepancy between the simulated two-dimensional (2D) acoustic model and the real volumetric acoustic scenario yields a noticeable interference leakage. This issue is currently coping our research efforts. This work corresponds to the final outcome of a BSc thesis accomplished at the academic institution of the authors.

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